

Fig. 4: An actual-size, single-side PCB for the electronic card lock using microcontroller

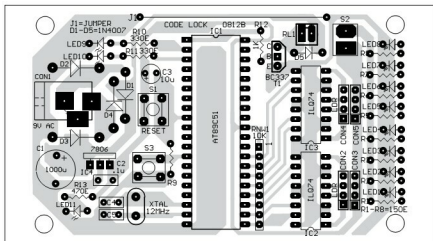


Fig. 5: Component layout for the PCB

optocoupler ILQ74.

The ILQ74 is a quad optically-coupled pair with a gallium-arsenide infrared LED and a silicon npn photo transistor. Switches S2 and S3 are used to provide power to the LEDs for glowing and interrupt signal to the microcontroller.

Microcontroller AT89C51 is the heart of the card lock. It is an 8-bit microcontroller with 4 kB of flash, 128 bytes of RAM, 32 input/output (I/O) lines, two 16-bit timers/counters, six interrupt sources, a full-duplex serial port, on-chip oscillator and clock circuitry.

Power-on reset is provided by the combination of resistor R9 and capacitor C3. Switch S1 is used for manual reset. A 12MHz crystal along with two 22pF capacitors provides the basic clock frequency to microcontroller AT89C51.

Port pins P2.0 through P2.7 of the microcontroller are pulled high with resistor network RNW1, which is further connected to the collectors of optocouplers IC2 and IC3, respectively. Port pins P1.0 through P1.2 are used to drive the relay, green LED and red LED, respectively.

The outputs of the optocouplers are fed to the microcontroller to compare the code. If the code matches with the code stored in the microcontroller, port pin P1.0 of AT89C51 goes high to drive relay RL1. At the same time P1.1 also goes high, driving the green LED (LED9) to indicate successful operation. Diode D5 acts as a freewheeling diode for relay RL1. Glowing of the red LED (LED10) indicates that the code does not match.

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